F70LA Life Insurance Mathematics A (2024/25)
Tutorial 7 (there is no Tutorial 6)

1. Why do insurers generally require evidence of health from a person applying for life insurance but not for a life annuity?

2. Let Y be the present value random variable of an n-year deferred annuity payable continuously at a rate of £1 per year to (x).

   (a) Explain how the expression

   $$Y = \bar{a}_{\overline{T_x}|} - \bar{a}_{\overline{\min\{n,T_x\}}|}$$

   holds.

   (b) Hence, or otherwise, show that

   $$\mathbb{E}(Y) = \frac{\bar{A}_{x:\overline{n}|} - \bar{A}_x}{\delta}.$$

3. Describe in words the benefits with the given PVs and write down an expression in terms of actuarial functions for the EPV.

   (a)

   $$Y_1 = \begin{cases} 200\,\bar{a}_{\overline{T_x}|} & \text{if } T_x \le 20, \\ 200\,\bar{a}_{\overline{20}|} & \text{if } T_x > 20. \end{cases}$$

   (b)

   $$Y_2 = \begin{cases} 500\,a_{\overline{10}|} & \text{if } K_x = 1, 2, \ldots, 10, \\ 500\,a_{\overline{K_x}|} & \text{if } K_x = 11, 12, \ldots. \end{cases}$$

4. (a) Describe the annuity with the following PV random variable:

   $$Y = \begin{cases} v^{T_x}\,\bar{a}_{\overline{n-T_x}|} & \text{if } T_x < n, \\ 0 & \text{if } T_x \ge n. \end{cases}$$

   The annuity is called a *Family Income Benefit*.

   (b) Show that $\mathbb{E}(Y) = \bar{a}_{\overline{n}|} - \bar{a}_{x:\overline{n}|}$.

   (c) Explain the answer in part (b) by general reasoning.

5. Given that $\ddot{a}_{40:\overline{15}|} = 9.615$, $a_{40:\overline{15}|} = 9.000$ and ${}_{15}p_{40} = 0.9227$, calculate the annual effective rate of interest used to calculate the annuity present values.

6. Suppose the annual effective rate of interest is $i = 0.04$. Valued at this interest rate and on the same mortality basis, $a_{50} = 12.545$, $a_{51} = 12.520$ and $a_{52} = 12.620$. Calculate ${}_2p_{50}$.

7. Given
$$ {}_{10}|\ddot{a}_x = 4.000, \quad \ddot{a}_x = 10.000, \quad v^{10}\,{}_{10}p_x = 0.375, \quad v = 0.94, $$
calculate $A^1_{x:\overline{10}|}$.

8. You are given the following select life table. Assuming an interest rate of 3% per annum effective, calculate

x	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
50	100 557	100 455	100 320	52
51	100 401	100 284	100 134	53
52	100 221	100 095	99 927	54
53	100 020	99 882	99 693	55
54	99 795	99 639	99 429	56

   (a) $\ddot{a}_{[50]:\overline{4}|}$.

   (b) $a_{[50]+1:\overline{4}|}$.

   (c) $(Ia)_{[50]:\overline{4}|}$.

   (d) $(IA)_{[50]:\overline{4}|}$.

   (e) The probability that the present value of an annuity-due of 1 unit per year issued to a select life age 50 years is less than 5.0.

Numerical solutions: Q5 6.000% per annum; Q6 0.92255; Q7 0.2650; Q8(a) 3.82156; Q8(b) 3.70123; Q8(c) 9.11620; Q8(d) 3.54759; Q8(e) 0.00859.